Low-frequency variant detection by ctDNA WES (TAV16) for minimal residual disease assessment and relapse prediction in resected PDAC

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1 Background

- Resected localized **pancreatic ductal adenocarcinoma (PDAC)** relapses frequently, driven by **minimal residual disease (MRD)**: tumor remnants that persist after surgery but escape detection by imaging.
- **Circulating tumor DNA** (ctDNA), the tumor-derived fraction of cell-free DNA (cfDNA), can reveal MRD in plasma.
- Challenge: ctDNA is heavily diluted, so informative variants sit at **very low variant allele frequency (VAF)**.
- Plasma **whole-exome sequencing (WES)** offers a tumor-agnostic way to find somatic variants.

AIM

Test whether **TAV16** (a ctDNA WES pipeline with patient-specific variant prioritization, validated in colorectal cancer[1]) detects low-frequency variants and predicts relapse in resected localized PDAC.

2 Methods

- **31 patients** with resected localized PDAC; plasma at **diagnosis** and **4–8 weeks after surgery**.
- cfDNA WES with **unique molecular identifiers (UMIs)**; paired white blood cells (WBC) to filter germline and clonal hematopoiesis (CHIP) variants.
- Somatic calling: Mutect2, LoFreq, VarScan. Germline calling: HaplotypeCaller.
- **TAV16** keeps the 16 highest-VAF baseline variants. **ctDNA+ = ≥1 variant persisting after surgery**.
- Kaplan-Meier and multivariable Cox models (age, sex, stage) for relapse-free (RFS) and overall survival (OS).

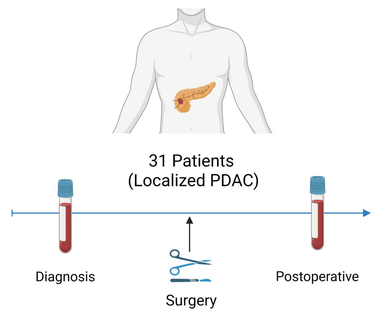


Figure 1. Study design: plasma sampled at diagnosis and 4–8 weeks after surgery.

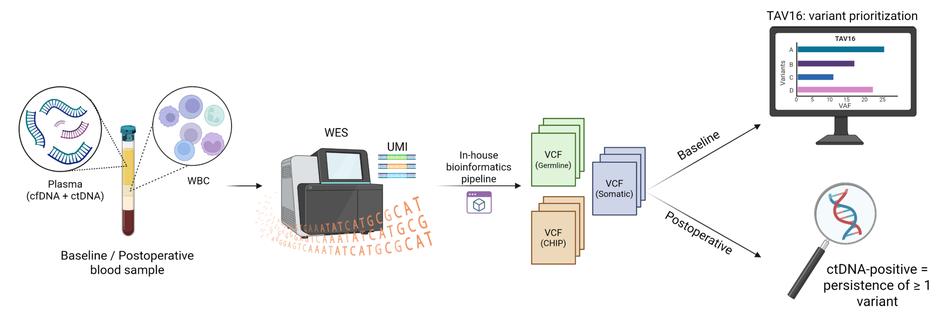


Figure 2. TAV16 workflow: plasma and WBC WES with UMIs; the top-16 baseline variants are tracked after surgery.

3 Results

Deep plasma coverage

Plasma reached ~600x median coverage; paired WBC (91x) were used to filter germline and CHIP variants.

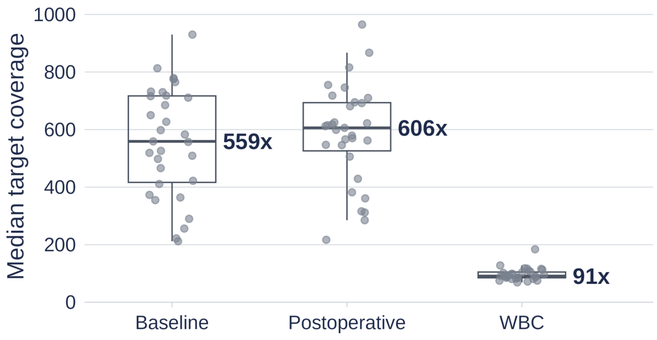


Figure 3. Median target coverage: 559x baseline and 606x postoperative plasma; 91x WBC.

79% of variants below 1% VAF

Tracked variants sat at very low VAF (median 0.55%; 79% below 1%), a range that challenges standard variant calling.

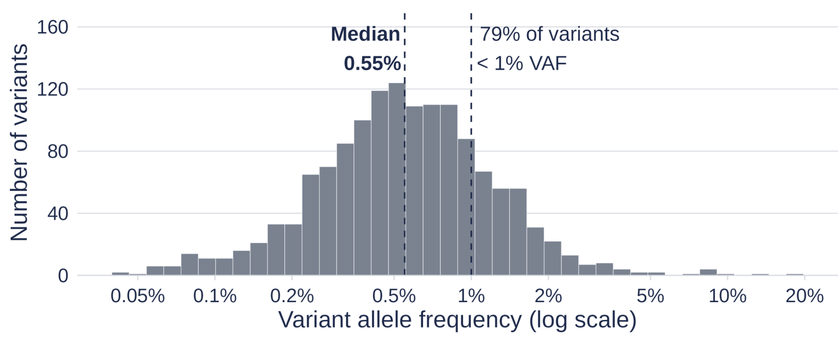


Figure 4. VAF of 1,410 oncogenic variant observations (31 patients, all timepoints); median 0.55%, 79% below 1%.

ctDNA+ after surgery anticipates relapse

19/20 ctDNA+ patients (95%) relapsed vs 4/11 ctDNA− (36%): **83% sensitivity and 88% specificity** for relapse.

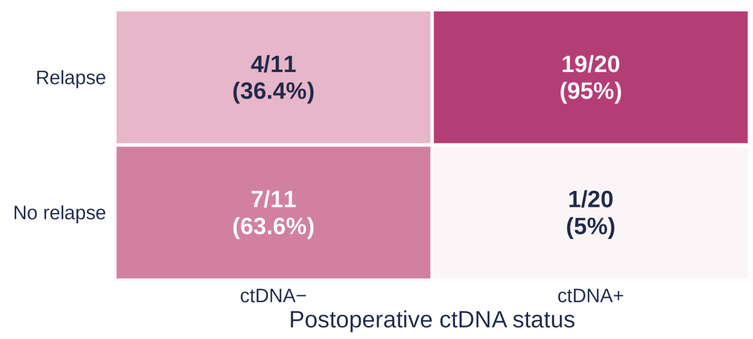


Figure 5. Relapse by postoperative ctDNA status; each cell shows n/N and % within its ctDNA group. Sensitivity: 19/23 relapsing patients were ctDNA+; specificity: 7/8 disease-free patients were ctDNA−. Fisher's exact test p = 0.00086.

ctDNA+ patients relapse earlier

After adjusting for age, sex and stage, postoperative ctDNA positivity remained an **independent predictor of RFS** (HR 6.11, 95% CI 1.95–19.14; p = 0.002).

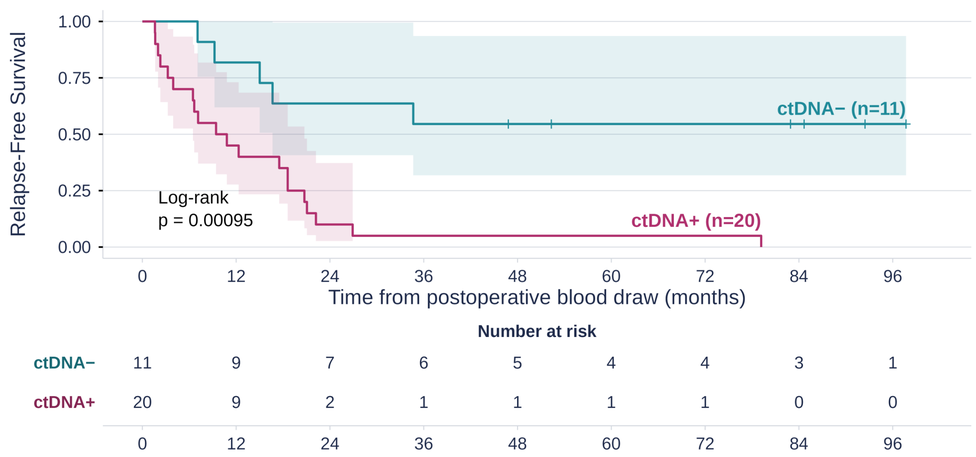


Figure 6. Relapse-free survival by postoperative ctDNA status: Kaplan-Meier curves (log-rank p = 0.00095) and multivariable Cox model adjusted for age, sex and stage. * p < 0.05.

…and have shorter overall survival

In the same multivariable model, ctDNA positivity was also **independently associated with shorter OS** (HR 8.65, 95% CI 2.19–34.14; p = 0.002).

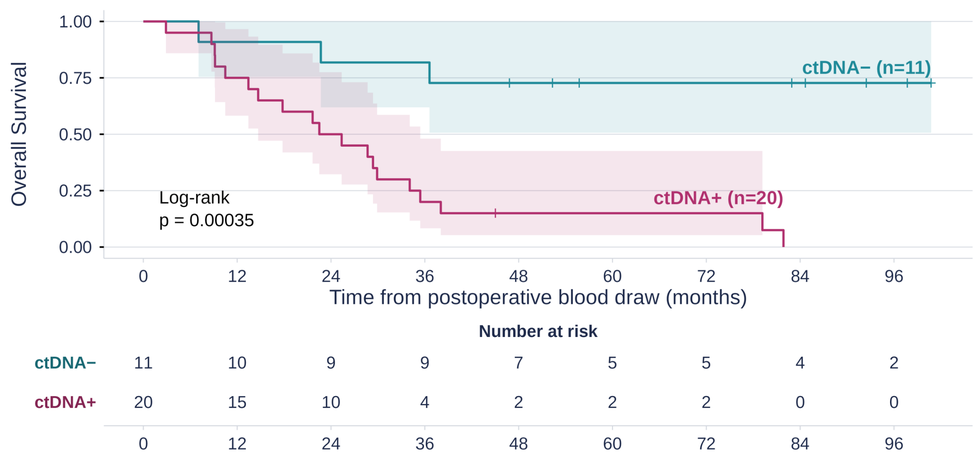


Figure 7. Overall survival by postoperative ctDNA status: Kaplan-Meier curves (log-rank p = 0.00035) and multivariable Cox model adjusted for age, sex and stage. * p < 0.05.

4 Conclusions

Plasma WES with UMIs and patient-specific variant prioritization (TAV16) captured tumor-derived variants below 1% VAF in resected localized PDAC, providing a tumor-agnostic postoperative MRD readout that independently predicted relapse and survival.

- TAV16 detects tumor-derived variants at **very low VAF** in postoperative plasma.
- Postoperative ctDNA positivity flags patients at **high risk of relapse** (83% sensitivity, 88% specificity).
- TAV16 **independently predicts RFS and OS**, supporting MRD-guided postoperative risk stratification.

[QR]

References 1. Martín-Arana J, Gimeno-Valiente F, Henriksen TV, et al. Whole-exome tumor-agnostic ctDNA analysis enhances minimal residual disease detection and reveals relapse mechanisms in localized colon cancer. *Nat Cancer*. 2025;6:1000–1016. doi:10.1038/s43018-025-00960-z

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